

Moondance

User Guide

A walkthrough of the tabs, menus, and actions.

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1. How Moondance is Organized

Moondance has six tabs along the bottom of the screen. Each tab is a different way of working with the same underlying data: your targets, collections, sessions, and observing settings.

- **At a Glance** — the home page. Tonight's moon, sun times, your settings summary, and a rotating astronomy quote.
- **Sessions** — your tonight's-plan workspace. Pick up to four targets and run the Moondance visibility chart.
- **Collections** — your wishlists and themed groupings. Folders, drag-reorder, and seeded collections live here.
- **Sky Map** — a dome view of tonight's sky. Filter pills, time scrubbing, and constellation drill-downs.
- **Search** — find any deep-sky object by name, catalog ID, or constellation.
- **Settings** — location, profiles, visibility filters, sound, and the export/import for backups.

Two important concepts

Sessions vs. Collections

A **Collection** is a curated wishlist that lives forever. A **Session** is a planning unit for a specific night — you can switch between sessions, snapshot their picks to History, and delete them when done. The default **Main** session is always present and can't be deleted.

Pick vs. Queue

Inside a session, your **queue** is the holding pen — every target you've added. Your **picks** are the (up to four) starred targets that feed the Moondance chart. Picking and unpicking doesn't remove from the queue.

2. At a Glance Tab

The home page. Reads top-to-bottom; nothing requires a tap to see the essentials. The header reads **MOONDANCE** with the tagline “Build Collections to Image Methodically”.

Moon block (top)

A small thumbnail of tonight's moon, the phase name (e.g., “Waning Gibbous”), and percent illumination.

Date and night times

- **SUN** — AD↓ (astronomical dusk, when the sky is fully dark) and AD↑ (astronomical dawn).
- **MOON** — moonrise and moonset times for the night.

Settings summary

A read-only block showing your active **Horizon** profile, **Moon** profile, current visibility **Filters**, and the queue contents of your active **Session**. Each row taps through to the Settings tab.

Quote

A daily-rotating astronomy quote. Includes Sagan, Hawking, Tyson, Pete Wingard, and a few Moondance-original lines.

Results card

Pinned card showing how many of your captured targets you've rated. Tap to jump to your rated images.

Tab bar (bottom)

All six tabs. The active tab shows in accent color.

3. Sessions Tab

Your planning workspace. The top of the screen shows the current session name with a chevron; below it, the queue of targets you added; at the bottom, the Moondance button.

Session picker (chevron menu)

Tap the session name to open the menu:

- **List of sessions** — checkmark next to the active one. Tap any session to switch.
- **New Session** — prompts for a name; creates an empty session and switches to it.
- **History (n)** — only shown when the active session has snapshots. Lists past Moondance runs; tap one to restore those picks (the current picks are snapshotted automatically before replacement).
- **Delete Session** — only shown when the active session is not the default Main. Removes the session and its history; an undo prompt is offered.

Status line

Reads “X of Y picked · Z in queue”. Picks are the starred chart subset (max 4); queue is everything in the session.

Profile names + Opportunity card

Shows the active Horizon and Moon profiles, plus a per-session opportunity score derived from the average of all queued targets.

Queue rows

Each row shows a thumbnail, the target name, type, constellation, tonight's rise/set, brightness, size, angular separation from the moon, opportunity tier, and a visibility bar.

Star (left)

Tap to pick the target for the Moondance chart. Up to four picks. When picked, a colored stripe appears on the row's leading edge matching the chart line color.

Tap row body

Opens the target's detail page (RA/Dec, Wikipedia, AstroBin, etc.).

? next to Opportunity

Opens an alert explaining the tier ranges (Ultra Rare under 30 hrs/yr, Rare 30–99, Uncommon 100–299, Seasonal 300–599, Common 600–999, Abundant 1000+).

Actions menu

Same menu offered everywhere ZebraRow renders:

- **Pick / Unpick for chart** — toggles the chart star.
- **Add to active Session** — promotes the target into the active session (no-op if already there).
- **Add to Collection...** — opens a sheet to drop into any collection (or create one).
- **Add Image** — opens the Image page where you can attach your own viewer-copy.
- **Add / Edit Note** — opens the note editor (1000 char cap).
- **Mark Captured / Not Captured** — toggles the capture flag.
- **Rate...** — submenu for Great, Okay, Redo, or Clear rating. Disabled until captured.
- **Moondance preview** — opens a single-target visibility chart.
- **Set moon filter...** — sessions only. Per-target broadband / narrowband override.
- **View on Wikipedia / View on AstroBin** — research links.
- **Remove from list** — deletes the target from the active session's queue (and unpicks if picked).

Visibility bar

The thin bar at the bottom of the row shows the dusk→dawn window; the colored segment is when the target is above your horizon profile.

Moondance button (bottom)

Greyed out until at least one target is picked and a location is set. Tap to compute night-by-night visibility for your picks across the full date range. On iPhone, the chart shows in landscape (rotate the phone). On iPad, the chart shows in place.

4. Collections Tab

Your wishlists, organized in folders if you want. Folders hold collections; collections hold targets. Sessions don't appear here — they live on the Sessions tab.

Top toolbar

- **Reorder** (top-left) — toggles drag-reorder mode for the top-level rows. Tap again to commit (label flips to **Done**).
- **+** (top-right) — opens the New menu: **New Collection** or **New Folder**.

List rows

Folders show a stacked-square icon; collections show a single folder. Each row has the name, an item count, and a one-line target preview.

Tap row

Opens the detail view for that collection or folder.

Right-side ellipsis (⋮) menu

Per-row actions:

- **Move into folder...** — submenu listing folders, top-level option, and “Create new folder...”.
- **Rename** — inline edit alert.
- **Delete** — confirmation alert; collection contents and members removed.

Collection detail view

Lists every target in the collection in priority order. Each row is a ZebraRow with the same actions menu and detail tap as elsewhere.

Reorder

Tap **Reorder** in the top-left to drag rows; **Done** commits.

Pick toggle

Inside the Main session collection, the leading icon is a star (picks for the Moondance chart). Inside any other collection, the leading icon is a circle/check that adds the target to the active session's queue.

Swipe left on a row

Reveals **Remove** — deletes the target from this collection.

Folder detail view

Lists child collections only (folders never contain targets directly). Add Collection at the top opens a menu to either move an existing collection here or create a new one nested inside.

5. Sky Map Tab

A dome view showing tonight's sky from your location. Shows constellations as pins with target counts, the moon, the night arc, and a draggable clock hand.

Filter pills (top, wraps to multiple lines)

- **All** — every target.
- **Session: <name>** — only the chart-picked targets in your active session.
- **Collections** — every target in any of your collections.
- **Galaxy / Nebula / Open Cluster / etc.** — by object type. Pulled from the catalog automatically.
- **Messier <captured>/108** — Messier-only with progress count.

Date selector

Below the pills. Left/right arrows step the observing date by one day. Tap the centered date to fast-jump back to tonight.

Dome map

Layout

- **Cardinal labels** — N, S, E, W around the rim.
- **Tick marks** — every 20° around the rim for orientation.
- **Altitude rings** — at 30° and 60° (faint). The rim itself is the horizon.
- **AD↓ / AD↑ markers** — astronomical dusk and dawn anchored on the night arc near the bottom.
- **Time pill** — center of the dome; updates as you scrub the clock.

Constellation pins

Each constellation that has visible targets shows a pin with a count badge (number of targets matching the active filter). Pins above the user's clearance are full-color; pins below are dimmed.

Tap a constellation pin

Opens the constellation detail sheet — a milky-way image with the constellation outlined, a small target locator that walks you through each target in turn, filter pills for type, and a scrollable list of targets grouped by type below.

Drag the clock hand

Scrubs time across the night. All constellation positions, the moon, and the time pill update live.

Bottom-right counter

Reads “N/88” — N = constellations above your horizon clearance, 88 = total. Tap to open the Constellation Status List.

Constellation Status List

A grouped list: **Above Clearance** (you can image), **Above Horizon but Below Clearance** (technically up but blocked by your trees/buildings/profile), **Below Horizon** (not up at all). Tap any row to drill into that constellation.

Top-left toolbar

Sky Map / All Constellations toggle — switches the body between the dome view and a flat alphabetical constellation list.

6. Search Tab

Find any target in the catalog by name, catalog ID, or constellation. Faster than scrolling Sky Map when you know what you're looking for.

Search bar

Free-text input. Matches against target names (“Andromeda”, “Ring Nebula”), catalog IDs (“M31”, “NGC 7000”, “IC 5068”, “Sh2-101”), and constellation names. Case-insensitive.

Type filter chips

Below the search bar — narrow the result list to a specific object type (Galaxy, Nebula, Open Cluster, etc.).

Result rows

ZebraRows with the standard kit: thumbnail, capture badge, type/constellation, visibility info, brightness, size, memberships, capture date, opportunity capsule, Actions menu, and tap-to-detail.

Note: Search bypasses your global Visibility Filters (max magnitude, min size, hide unknown). It always shows the full catalog so a specific lookup never fails.

7. Settings Tab

Everything that controls how Moondance plans for you. The big sections:

Location

- **Preset list** — major US/world cities with lat/lon/timezone built in.
- **Search** — type a city name; resolves via the bundled location service.
- **Custom Location** — toggle on; enter latitude, longitude, elevation, and timezone manually. Used as-is for all astronomy calculations.

Date Range

Discrete slider 30–365 days. Controls how far the Moondance chart looks ahead.

Visibility Filters

Three settings that hide noise on browse views (Sky Map, Collections, Search-by-type). Explicit-name search ignores them.

- **Max display magnitude** — hide objects dimmer than this.
- **Min display size (arcmin)** — hide objects smaller than this. 0 = no minimum.
- **Hide unknown magnitude** — drops Sh2/LDN/dark nebulae that don't have a numeric magnitude.
- **Max chart targets** — picks-cap (1–4) for the Moondance chart. Default 4.

Horizon Profiles

Each profile is eight directional altitude minimums (N, NE, E, SE, S, SW, W, NW). “Clearance” on rows uses the profile to decide whether a target rises above your trees/buildings/hills in the right direction.

- **Edit profile** — sliders per direction.
- **Add profile** — duplicate the active one with a new name; switch profiles per night/site.
- **Active** — pick any profile to make active.

Moon Profiles

Each profile defines minimum target/moon angular separation by moon phase tier (e.g., “new moon → 0°,” “full moon → 60°”). Used to color the Moondance chart moon-up bars.

- **Edit / Add / Switch** — same shape as horizon profiles.
- **Per-target overrides** — sessions only. The Set moon filter... menu item on a row picks broadband/narrowband for that one target without changing the active profile.

Sound Effects

Toggle for tap, capture, and removal sounds.

Quick Start

Reopens the in-app first-launch walkthrough.

Walkthrough Videos

Opens sidestepstudio.com/moondance/quickstart.html in Safari — the external hub for video walkthroughs and the PDF user guide.

Export & Import Settings & Data

- **Export** — produces a single JSON file containing every collection, session, folder, capture, note, history snapshot, thumbnail, and user-uploaded image. Share via the system share sheet (AirDrop, save to Files, email, etc.).
- **Import** — pick a previously exported JSON. Replaces every key in the export. Recommended for migrating to a new device, restoring after a wipe, or moving from Dev to production.

Note: Export files are sized in MB (not KB) when thumbnails and user images are present. A small file means image data isn't bundled.

Debug (DEBUG builds only)

Replay First-Launch Experience, memory profile probes, scoring test sets, and other developer tools. Hidden in App Store builds.

8. Target Detail Page

Reached by tapping the body of any ZebraRow. A full read-out of everything Moondance knows about a target.

Header

Target name, type, constellation, opportunity tier capsule + ? for the tier explanation.

Tonight

- **Rise / Set** — when the target clears your horizon profile and when it sets again.
- **Transit altitude** — highest altitude reached tonight.
- **Best date this year** — the night where conditions hit their peak.
- **Moon separation** — effective (90°-folded) angular separation.

Coordinates & Brightness

- **RA / Dec** — both decimal-degrees and HMS/DMS forms.
- **Magnitude / Surface Brightness** — when present in the catalog.
- **Sh2 brightness class** — Bright / Medium / Faint, when applicable. ? button explains.
- **LDN opacity class** — 1–6 for Lynds Dark Nebulae. ? button explains.

Charts (button)

Opens a per-target ChartsSheet with night-by-night altitude over your full date range — useful for picking the best week to shoot.

Image (button)

Opens the Image page (next section).

Notes / Capture / Rating

Inline editor for the per-target note (1000 char cap), Captured toggle, and rating selector (Great / Okay / Redo / Clear).

Memberships

Lists every session and collection this target is in. Sessions render in accent color so they stand out from regular collections.

9. Image Page

Where you attach a viewer-copy image to a target. Reached from the **Image** button on the target detail page or the **Add Image** action in any row's Actions menu.

Image preview

Shows your attached image when present, otherwise a placeholder.

Wikipedia summary

Auto-loaded extract for the target. Falls back to a Wikipedia search link if no direct article matches.

Action links

- **View on Wikipedia** — opens the article (or search) in Safari.
- **View on AstroBin** — searches AstroBin for the target name, sorted by likes.
- **Search the Web** — Google search for “<name>” astronomy. Often the fastest way to find annotated images that show *which* part of a wide field is the actual object.

My Image

- **This is my image** — checkbox; flags the photo as your own. Yellow star badge appears on every row showing this target.
- **Captured** — toggle. Marks the target as imaged.
- **Rating** — Great / Okay / Redo / clear, available after captured is on.
- **Wiki** button — pulls Wikipedia's reference image as a placeholder when you haven't attached your own. Falls back to the DESI Legacy Survey if Wikipedia has no image.

10. Moondance Chart (Landscape)

Reached by tapping **Moondance** on the Sessions tab. On iPhone, rotate to landscape to view; on iPad, the chart renders in place.

What it shows

A bar chart with one row per night across your date range. Each night shows colored bars for each picked target indicating when they're up and the imaging quality (moon brightness, phase, separation, altitude) at each hour.

Color key

- **Green** — perfect imaging conditions (moon down or far enough away).
- **Yellow** — allowable: moon is up but separation meets your moon profile.
- **Orange / Red** — poor: moon too bright or too close.
- **Grey** — target below clearance for that hour.

Tap a night

Opens a per-night detail popup with sun/moon times, percent moon, effective separation per target, and a verdict (“Imageable” vs. “Not imageable — moon too bright”, etc.).

Back / Done

Top-left back button (or rotate back to portrait) returns to the Sessions tab.

11. Capture and Rating System

Tracks which targets you've imaged and how well.

Mark Captured

From any Actions menu, or via the toggle on the Image page. A capture record stores the target ID, date, and (optional) rating.

Rating

- **Great** — yellow star badge on every row.
- **Okay** — checkmark badge.
- **Redo** — circular-arrow badge; reminds you the image needs reshooting.
- Captured-without-rating shows a plain yellow star (no glyph).

Messier tracker

Automatic tally — every Messier you mark Captured updates the Sky Map's **Messier X/108** pill.

My Image star

Independent flag (set on the Image page) that says “this viewer-copy image is mine, not a survey reference.” Renders as a yellow star in the row's inline glyph.

Captured filter

In Sky Map and any seeded-collection run, you can opt to **exclude already captured** so suggestions surface only fresh work.

Capture date

Right-aligned in muted yellow on every row showing a captured target. Format: YY/M/D.

12. Notes

Free-form per-target text. Up to 1000 characters.

Add / Edit

From any Actions menu pick **Add Note** or **Edit Note**. Opens a sheet with a multi-line editor and a character counter.

Display

- In sessions: the first line is shown inline on the row, with a small note icon.
- Elsewhere: a **Note** link appears on the row; tap to popover the full text.

Use cases

Gear settings (“use 0.95x reducer”), processing reminders (“don’t forget continuum subtraction”), first-light memories, or just a one-line caption.

13. Backup and Restore

Settings → Export Settings & Data produces a single JSON file containing your complete app state. Import is the inverse.

What's included

- Location, custom coordinates, timezone, date range.
- All horizon profiles, moon profiles.
- Visibility filters, max chart targets.
- Every collection, folder, and session (including drag-reorder priority).
- Every capture record (including date and rating).
- Every note (full text).
- Every history snapshot (per-session).
- Messier tracker state.
- Active session ID.
- All thumbnails (JPEG, base64-encoded).
- All viewer-copy user images (JPEG, base64-encoded).

Why the file is large

A backup with thumbnails and user images is megabytes, not kilobytes. A small file (under 100 KB) means image data isn't bundled — re-export.

Recommended cadence

Export after every significant editing session and store in two places (e.g., AirDrop to your Mac and email to yourself, or save to iCloud Drive in two folders).

Restoring

On the destination device: Settings → Import Settings & Data → pick the JSON. Every key in the file replaces the matching key on the device. Targets, captures, notes, thumbnails, and user images all return as they were.

Note: On a fresh install of Moondance from TestFlight or the App Store, the binary alone won't restore your data — you need the JSON. iCloud Backup of the device sandbox is a separate, iOS-managed safety net that captures the same data nightly.

14. Conveniences and Tips

Shake to undo

Shake the phone to undo the last destructive action — restores deleted collection members, deleted collections, deleted folders, and history snapshots. Works for several seconds after the action.

Tab persistence

Moondance remembers which tab you were on between launches.

Tonight button

On the Sky Map's date selector, tap the centered date to fast-jump back to today's observing date from any other date.

Dawn-aware date

After astronomical dawn but before noon, Moondance still treats “tonight” as the previous evening's night, so the AD↑ / AD↓ times you see at 5 AM still describe the night you were just observing.

Dark mode by design

The whole app is built dark — there is no light mode. Designed for night-time use without blowing out your night vision.

Per-target moon-filter override

From a session row's Actions menu, **Set moon filter...** lets you switch a single target between broadband and narrowband for the chart's color logic — without changing the active moon profile.

Membership tags

Every row shows the sessions and collections it belongs to. Sessions are rendered in accent purple; collections in muted gray. Useful for spotting double-booked targets across your plans.

Quick targeting via constellation pill

In the constellation drill-down list, the small locator photo at the top has a left/right arrow to walk through each target one at a time — fast review of an entire constellation's contents.